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Muscle hypertrophy response across four muscles involved in the bench press exercise: Randomized 10 weeks training intervention

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ABSTRACT

Background and purpose: Resistance training exercise provides increases in muscle size and is used by coaches and health care professional as a tool to improve functional performance. The aim of the present study was to investigate the effect of 10 weeks of resistance training program performed on the bench press (BP) exercise on the hypertrophic responses of four different muscles (pectoralis major, anterior and medial deltoid, brachii, and pectoralis minor) involved in the task compared to controls.
Methods: Twenty-four healthy males were recruited, and thirteen performed a resistance training intervention while eleven were control. RT group trained for 10 weeks and the protocol consisted of a time under tension for each set of 36s (3–4 sets) with 12 repetitions with an intensity of 50–55% of the 1RM, a training frequency of 3 times a week, with a 3 min rest between sets. Muscle cross-sectional area (CSA) was measure by magnetic resonance imaging.
Results: Individuals in the RT group demonstrated higher changes in CSA of the pectoralis major, pectoralis minor, anterior deltoid, and triceps brachii ($P \leq 0.019$) than in the Control group. It was identified in the RT group higher increases in CSA of all muscles compared to medial deltoid ($P \leq 0.016$), while pectoralis major demonstrated larger increases in CSA than pectoralis minor and triceps brachii ($P \leq 0.030$).
Conclusions: We demonstrated that 10 weeks of resistance training performed on the BP exercise led to increases in muscle size of the muscles involved in the task, but not in the same magnitude.

1. Introduction

Perform resistance training exercise has shown to provide increases in muscle size (muscle hypertrophy) (Folland and Williams, 2007). Hypertrophy can be noticed after resistance training interventions executed on the bench press (BP) exercise (Martins-Costa et al., 2021; Pareja-Blanco et al., 2020; Yasuda et al., 2011). The BP is a very popular multi-joint exercise (i.e., shoulder, elbow, and wrist joint) that involves many muscles to execute it. The research studies that investigated the effect of a BP training intervention on muscle size have limited the analysis to one or two prime movers (e.g., pectoralis major and triceps brachii) (Brandão et al., 2020; Larsen et al., 2021; Martins-Costa et al., 2021; Pareja-Blanco et al., 2020; Yasuda et al., 2011). However, other muscles involved in the BP may also present hypertrophy after a period of training, as demonstrated elsewhere in other multiple joint exercises (e.g., squat) (Kubo et al., 2019). Understanding if and how much other

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muscles increases the size after a training performed on the BP will provide further insight and possibilities for exercise prescription among coaches and health care professionals.

During the BP exercise, the pectoralis major, anterior deltoid and triceps brachii (Larsen et al., 2021) have often their muscle activity measured by electromyography (Stastny et al., 2017). Pectoralis major and triceps brachii were the most investigated muscles when researchers were assessing muscle size pre-to a post-training intervention targeting to verify hypertrophy using BP exercise (Martins-Costa et al., 2021; Ogasawara et al., 2013; Pareja-Blanco et al., 2020; Yasuda et al., 2011). Although the anterior deltoid is also a prime mover in the horizontal adduction of the shoulder during the BP exercise (Muscolino, 2017; Stastny et al., 2017), no studies seem to have analyzed its hypertrophic response (Muscolino, 2017; Stastny et al., 2017). Therefore, understanding the effect of resistance training on the hypertrophic response of this portion of the deltoid will help to understand which muscles are affected by the bench press exercise. It is also noteworthy that some studies have shown interest in measuring muscle activation of the medial portion of the deltoid muscle, possibly due to its arm stabilizing role during the bench press exercise (Campos et al., 2020; Schick et al., 2010). However, to the best of our knowledge, no study has analyzed the chronic effect of training on the muscle size of the medial portion of the deltoid. If the bench press exercise also causes an increase in this portion of the deltoid muscle, coaches and health care professionals should consider this additive effect when designing resistance training programs. For instance, this may lead to a reduction in the number of exercises focusing in this specific muscle.

The pectoralis minor is another muscle involved in the BP exercise (Bhatia et al., 2007). Although the main actions of the pectoralis minor are related to the depression and anterior tilt of the scapula (Bhatia et al., 2007; Neumann, 2016), this muscle seems to collaborate with the protraction movement, as well as limiting the retraction of the scapula (Neumann, 2016). Depending on the BP exercise technique adopted, scapula protraction may occur at the end of the bar lifting phase (Borges et al., 2019; Hackett et al., 2014). It should be noted that changes in the length and stiffness of the pectoralis minor muscle have been associated with scapular dyskinesis (Kibler et al., 2013). Considering that the increase in muscle cross-sectional area seems to promote an increase in its stiffness (Blazevich, 2019; Ryan et al., 2009), and that resistance training can impact both the increase in muscle cross-sectional area and stiffness (Blazevich, 2019; Reeves et al., 2003), it would be important to understand the effect of training on BP in increasing the pectoralis minor muscle size. Although shortening and tendinopathy of the pectoralis minor have already been reported among practitioners of the BP exercise (Bhatia et al., 2007; Cutrufello et al., 2017), no studies were found that analyzed the chronic effect of BP training on the size of this muscle.

Therefore, the present study aimed to investigate the effect of 10 weeks of resistance training program performed on the BP exercise on the hypertrophic responses of different muscles (pectoralis major, anterior and medial deltoid, triceps brachii, and pectoralis minor) involved in the task compared to controls. Based on the previous research and muscle contribution to the task we hypothesized all muscles will present a hypertrophic response, but the pectoralis major, anterior deltoid, and triceps brachii muscles of the individuals in the training group will show larger increases in muscle size compared to control group. An exploratory (secondary) aim is to investigate within the group that performed the resistance training, if there are differences in increase in muscle size between muscles. We hypothesized that muscles that are prime movers in the task will show higher increases in muscles size than the others.

2. Methods

Twenty-six healthy males who were not involved in any activity involving strength training during the past 6 months and with no history of injuries in the wrist, elbow, or shoulder joints volunteer for this study. A total sample size of 24 subjects (12 per group) were determined by using GPower (version 3.0.1, Germany) based on a test using an analysis of variance (ANOVA) (repeated-measures, within-between interaction), with an alpha level of 0.05, power of 0.80; effect size (ES) of 0.30 (obtained from the triceps brachii muscle base on a previous study (Ogasawara et al., 2011)), number of groups, 2; number of measurements, 2; correlation among measures of 0.5; and a nonsphericity correction of 1. We recruited one participant more for each group (13) to account for a possible dropout. All participants then engaged in a familiarization session for all measurement procedures before completing the 1RM test on the bench press. The results of the 1RM test were then used as a classification parameter (matched-groups design), and the participants were subdivided and randomly assigned into one of the two groups (Control or Resistance Training [RT]). The matched-group design ensured that participants were evenly distributed between groups based on their 1RM values, preventing any imbalances of stronger or weaker individuals in one group that could skew the results. All participants provided written informed consent before their participation. This study was approved by the University's Institutional Review Board.

Participants attended the clinic for a magnetic resonance imaging (MRI) scan followed by two laboratory sessions (familiarization and main test session) separated by a minimum of 72 h and a maximum of 96 h. After that, the results of the 1RM test were then used as a classification parameter (matched-groups design), and the participants were subdivided into one of the 2 groups, Control or Resistance Trained (RT) groups. During 10 weeks, RT group performed a resistance training protocol (supervised sessions), whereas a Control did not perform any training. The Control group was asked to keep the normal routine during the study period. After the end of the training period, all participants performed new MRI assessments (72–96 h). Testing sessions were performed at a consistent time of the day and participants were instructed to avoid any exercise in the 48 h before each session. Additionally, participants were instructed to maintain the same dietary behaviour throughout the study.

Participants underwent an MRI examination to measure the CSA of the pectoralis major, pectoralis minor, triceps brachii, anterior deltoid, and medial deltoid muscles. Before the exam, participants were seated for 20 min and then placed in a supine position on a stretcher with their muscles relaxed for at least 15 min. MRI scans were performed on a Sigma HDX 1.5 T (GE Medical System, USA) with gradient-echo in phase/out phase technique T1-weighted, 120 ms repetition time, 2 ms echo time, a field of view of 400 mm, resolution of 320 x 224 pixels, and 8.0 mm slice thickness obtained continuously. During the examination, the upper limb was fixed in a normal anatomical position and parallel to the trunk and the volunteers were instructed not to move during image acquisition. Considering the existence of chest movement during breathing, the participants were also asked to perform apnea in some stages of the exam, to obtain a better quality of the images.

CSA of all muscles was identified by visual inspection and manually a line contoured the muscle for each slice (cm^2; Fig. 1) by using the MIPAV program (MIPAV 10.0.0, National Institutes of Health Center for Information Technology). All images were analyzed in the axial plane. To determine the CSA of the pectoralis major, the first image after the insertion of the pectoralis major into the humerus was used, which represented approximately 60% of the distance of the muscle in relation to the greater tubercle of the humerus (initial point) to the epicondyle of the elbow (final point) (Fig. 1A and B). A previous study (Yasuda et al., 2011) used an analysis point similar to that adopted in the present study, which facilitated the clear identification of the muscle belly of the pectoralis major in the axial plane. The same image was used to calculate the CSA of the pectoralis minor muscle. To determine CSA of the triceps brachii muscle, it was used as a reference for the distance between the greater tubercle of the humerus as the initial point while the lateral epicondyle of the humerus as the final point. After that, 60% of the total distance of the abovementioned anatomical reference was used to draw the muscle (Kanehisa et al., 2004) (Fig. 1E). To determine CSA

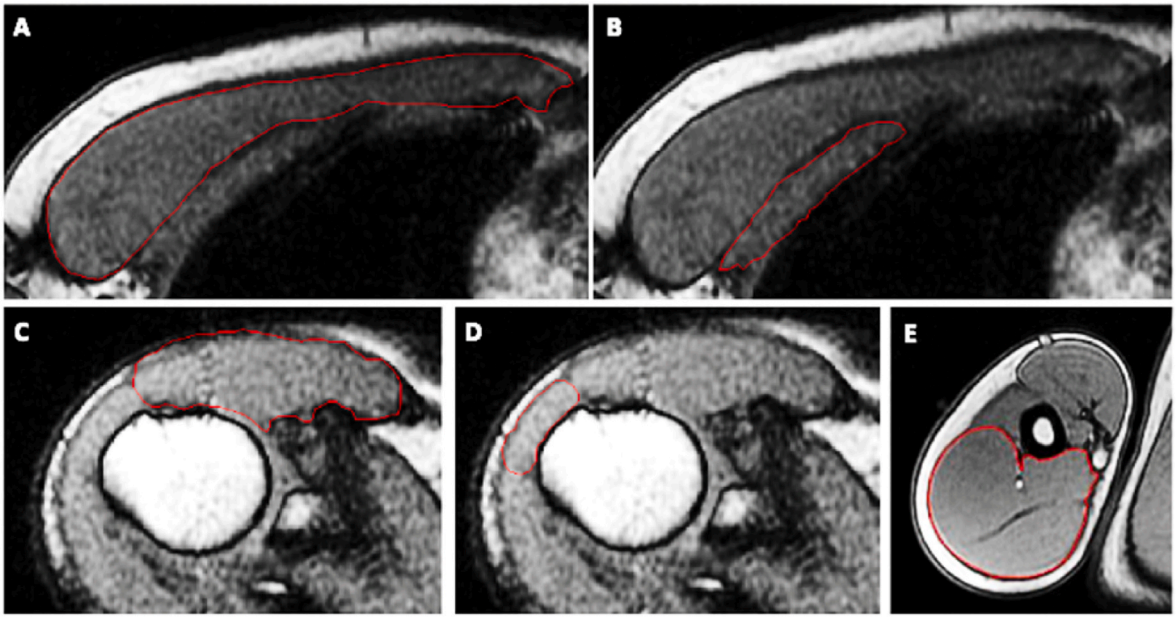

Fig. 1. Images of cross-sectional area drawing from one individual. Pectoralis major (A), pectoralis minor (B), anterior deltoid (C), medial deltoid (D), and triceps brachii (E).

of the anterior and medial deltoids, it was used as anatomical reference to the first appearance of the intertubercular sulcus of the humerus, and both muscles were drawn in the image (Fig. 1C and D).

The 1RM was performed in two baseline sessions and one post-training session (4–5 days after the last training session). The first session was considered a familiarization session. In all session participants were positioned on the equipment (Smith machine, MASTER Belo Horizonte, Brazil), with the position of the hands on the bar being standardized as twice the biacromial distance with the middle finger being the reference for marking (Yasuda et al., 2011). Additionally, to ensure the same positioning of the participant across all testing and training sessions, tape measures were fixed to the bar and the bench as a reference to replicate the individual location of the hands and head. The general guidelines for determining the performance of 1RM were as follows: (1) make a maximum of 6 attempts, with pauses of 3 min between each one, and (2) gradually increase the external resistance with each repetition (minimum difference of 2 kg per attempt). The external resistance on the bar was progressively increased until the volunteer was unable to perform the concentric action. At this point, the value of 1RM corresponded to the highest weight raised in the previous attempt. Same procedures were used elsewhere (Martins-Costa et al., 2016).

RT group trained for 10 weeks with all training sessions being supervised. The RT protocol consisted of a time under tension for each set of 36s with 12 repetitions [repetition duration of 3 s (1.5 s concentric and 1.5 eccentric muscle actions)], with an intensity of 50–55% of the 1RM, a training frequency of 3 times a week (48–72 h between sessions), 3 min rest between sets, and 3–4 sets number. The duration of the repetition was controlled by a metronome (M&MSystem, Hungen, Germany). Participants from the RT groups completed 30 training sessions. During the 10 weeks of training, when the participants were unable to attend one of the sessions in one week, the individual was allowed to perform training sessions 4 times a week in the following week, adopting a minimum break of 24 h between sessions. The protocol was based on previous research with a similar population that showed increases in muscle size (Tanimoto et al., 2008).

A Shapiro-Wilk test was used to assess the normality of the data. To identify possible baseline differences in CSA between groups a student t-test was performed for each muscle. A two-way ANOVA with a between factor (muscle x group) was performed by using the CSA percentual values [((CSA post-training - CSA pre-training)/CSA pre-training)*100], and LSD post-hoc was used when necessary. Effect size (ES) was calculated as previously detailed (Cohen, 1988) and classified as follows: <0.20 = "trivial"; 0.20–0.49 = "small"; 0.50–0.79 = "moderate"; or ≥ 0.80 = "large". Statistical analysis was performed using SPSS version 26 (IBM Corporation, Armonk, New York, USA). The significance level was set at $P \leq 0.05$.

3. Results

The Control group began with thirteen male participants, but due to dropouts (participants did not provide any specific reason for dropout), it concluded with eleven male participants, having an average age (mean ± SD) of 21.8 ± 5.0 years, and body mass, and height of 79.2 ± 12.2 Kg and 1.78 ± 0.08 m, respectively. The RT group (thirteen males) presented an average age of 24.3 ± 4.4 years, body mass of 73.4 ± 10.7 Kg, and height of 1.75 ± 0.05 m.

No baseline differences were found in the muscle CSA values across all analyzed muscles between the groups ($P \geq 0.207$), Table 1. There was a main effect of muscle ($F_{(4,88)} = 4.722$, $P = 0.002$), an interaction ($F_{(4,88)} = 4.227$, $P = 0.004$), and a main effect of group ($P < 0.001$). Individuals in the RT group demonstrated higher changes in CSA of the pectoralis major (19.9% vs. −3.6%; $P < 0.001$, ES = 2.3, "large"), pectoralis minor (11.5% vs −0.5%; $P = 0.019$, ES = 0.76, "moderate"), anterior deltoid (14.9% vs 3.7%; $P = 0.003$, ES = 1.26, "large"), and triceps brachii (11.5% vs −2.4%; $P = 0.003$, ES = 1.52, "large") than in the Control group, but no differences for medial deltoid (−0.9% vs −2.0; $P = 0.794$, ES = 0.21, "small"), Fig. 2.

When looking at the RT group, all muscles showed higher increases

Table 1
Values of muscle cross-sectional area (CSA) pre and post training intervention for Control and Resistance Training groups.

	Control		Resistance Training	
Muscles (CSA)	Pre	Post	Pre	Post
Pectoralis Major (cm^2)	28.2 ± 3.9	27.2 ± 3.8	29.2 ± 5.7	35.7 ± 7.2
Pectoralis Minor (cm^2)	4.9 ± 1.1	4.8 ± 1.1	4.9 ± 1.1	5.6 ± 1.9
Anterior Deltoid (cm^2)	12.7 ± 1.4	13.2 ± 1.9	11.6 ± 2.1	13.5 ± 2.4
Medial Deltoid (cm^2)	3.2 ± 0.7	3.1 ± 0.7	3.1 ± 1.3	3.1 ± 1.4
Triceps Brachii (cm^2)	25.3 ± 4.3	25.8 ± 4.0	26.2 ± 5.5	29.5 ± 6.1

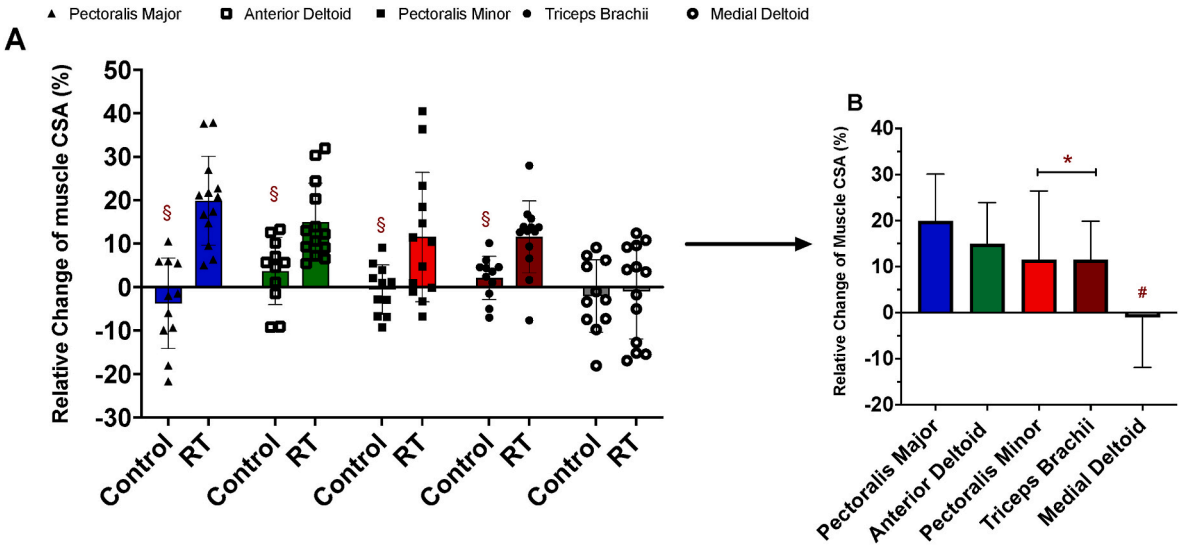

Fig. 2. (A) Percentage change in muscle cross-sectional area (CSA) with individual values for pectoralis major (closed triangle), anterior deltoid (open square), pectoralis minor (closed square), triceps brachii (closed circle), and medial deltoid (open circle) from pre to post training period. Symbol indicates: § difference between groups [Control vs. Resistance Trained (RT)]. (B) Percentage change in muscle CSA, only for the RT group. Symbol indicates: # difference between the medial deltoid muscle in relation to the other muscles; * difference between the triceps brachii and pectoralis minor in relation to the pectoralis major muscle.

in CSA than medial deltoid ($P \leq 0.016$, ES ≥ 0.52, "moderate" for anterior deltoid and triceps brachii, ES = 0.47, "small" for pectoralis major, and ES = 0.05 for pectoralis minor "trivial"). Additionally, the pectoralis major demonstrated larger increases in CSA than pectoralis minor ($P = 0.030$, ES = 0.76, "moderate") and triceps brachii ($P = 0.014$, ES = 1.60, "large"), but did not show difference when compared to the anterior deltoid ($P = 0.267$, ES = 1.28, "large"). However, no further differences were found for the other muscles.

4. Discussion

The present investigation explored the effect of 10 weeks resistance training program performed on the BP exercise on the hypertrophic responses of the different muscles involved in the task and compared to controls. As we hypothesized, pectoralis major, anterior deltoid, and triceps brachii muscles showed a greater increase in muscle size after the training intervention compared to controls. However, pectoralis minor also showed an increase in size after training. Additionally, the magnitude of the hypertrophic response was not the same for the evaluated muscles. The pectoralis major hypertrophied more than the other muscles, except for the anterior deltoid. The training intervention used in the present study might be used for coaches and health care professionals if the aim is to generate hypertrophy of the different muscles involved in BP investigated here.

We demonstrated a 19.9%, 14.9%, and a 11.5% increase in muscle size for the pectoralis major, anterior deltoid, and triceps brachii (respectively) in the training group after the intervention performed here. Similarly, another study (Yasuda et al., 2011) that performed BP training intervention for 10 weeks also showed hypertrophy in the pectoralis major (~10%) and triceps brachii (~5%), measured by MRI, values very close to those found by Brandão et al. (2020). Furthermore, a BP training intervention of 12 weeks demonstrated a similar increase in muscle size (measured by ultrasound) of the pectoralis major (~20%) and triceps brachii (12%) (Kikuchi and Nakazato, 2017). The different results between studies might be due to the different training configurations (e.g., training frequency, volume, and others) that are known to affect muscle hypertrophy (Figueiredo et al., 2018; Lacerda et al., 2019). To the best of the authors knowledge, this is the first study that investigated the hypertrophic responses of the anterior deltoid after a BP training intervention. This information strengthens the concept that the anterior deltoid has an important role during the BP exercise, even though studies based on the electromyographic response have suggested a lower adaptation potential of this muscle (Stastny et al., 2017). Thus, the BP training intervention performed here was capable to improve the muscle size of all the prime movers involved in the BP exercise and might be used when these muscles are the target of the training.

This is the first investigation to analyze the hypertrophic responses of the medial deltoid and pectoralis minor muscles after a training intervention on the BP exercise. It was demonstrated that the pectoralis minor increased 11.5% after the training period. Thus, considering this large increase in muscle size, it is reasonable to conclude that the pectoralis minor muscle might have an important role during the BP exercise, even taking into account that the participants of the present study were not instructed to perform scapular protraction at the end of the concentric phase of the movement. As previously presented, changes in the length and stiffness of the pectoralis minor muscle have been associated with scapular dyskinesia, due to its pull on the coracoid process, which could create anterior tilt of the scapula (Kibler et al., 2013). Considering the increase in CSA of the pectoralis minor muscle observed here, and the potential effect of resistance training on the change in muscle stiffness (Blazevich, 2019), future studies and health care professionals should consider the effect of bench press training on scapular kinematics or other outcomes associated with scapular dyskinesis.

Nonetheless, the medial deltoid showed no increases in muscle size after the training intervention. It is possible that muscles that stabilize the movement do not receive as much stimulus during the contraction as the other muscles evaluated in this study, hence this would decrease the mechanical or metabolic stimulus which might have an effect on the hypertrophic responses to the exercise (Wackerhage et al., 2019). Additionally, it should be noted that, in the present study, the smith machine bench press was used, which required less activation of the medial deltoid when compared to the barbell bench press (Schick et al., 2010). Although there is no evidence about an association between muscle activation and hypertrophic response, higher muscle activation may lead to higher metabolic stress which is related to muscle hypertrophic response (Wackerhage et al., 2019).

The present study has limitations and strengths that should be highlighted. The results we present here are restricted to the training configuration performed here; thus, using a different training structure may affect the outcome of the training (e.g., hypertrophic responses).

Our power analysis determined a requirement of 12 individuals per group. Although we initially started with 13 individuals, we concluded with only 11 in one group. Nonetheless, the effect sizes were large. In the present study, although we used a reference method for measuring muscle size (MRI), it was not possible to obtain CSA measurements in several regions of the different muscles. Additionally, the study was performed on the smith machine and, considering the expectation of differences in muscle activation between different ways of performing the bench press exercise (e.g., barbell bench press, dumbbell bench press), extrapolation of these results should be done carefully. We recommend that future studies explore the effect of different training configurations on the muscles involved in the BP exercise. The present investigation is not limiting the analysis to only a couple of muscles. We are providing robust evidence about the differences in muscle size of four different muscles involved in the BP exercise. Furthermore, our study sheds light on the ongoing debate among coaches and trainers regarding the muscle that undergoes the most substantial growth through the utilization of bench press exercises. While our study possesses a distinct training configuration, it provides valuable insights into this discussion. Therefore, coaches and health care professionals that need to increase the muscle size of any of the muscles investigated here can use the proposed training intervention to achieve their goal. Additionally, the present study highlights that even muscle that are not prime movers in an exercise might change with training.

5. Conclusion

In conclusion, the present study showed that 10 weeks of resistance training performed on the BP exercise increases muscle size of the prime movers (pectoralis major, anterior deltoid, and triceps brachii) and in a muscle with other function roles (pectoralis minor). More specifically, BP exercise promoted a greater hypertrophic response of the pectoralis major compared to the triceps brachii and pectoralis minor, but not different from the anterior deltoid. These findings should be considered when quantifying the training volume in multi-joint exercises (e.g., consider a smaller fraction of a set for the triceps brachii when prescribing the BP exercise) (Schoenfeld et al., 2019). Additionally, these results reinforce the importance of future research quantifying the hypertrophic response of different muscles that act in multi-joint exercises, whether they are primary motors or not.

CRediT authorship contribution statement

Marcel B. Lanza: Writing – review & editing, Writing – original draft, Visualization, Software, Formal analysis, Data curation. **Gustavo C. Prado:** Writing – review & editing, Writing – original draft, Software, Formal analysis. **Lucas T. Lacerda:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Ricardo Reis Dinardi:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Data curation. **Lúcio Honório Carvalho Junior:** Writing – review & editing, Writing – original draft, Formal analysis. **Rodrigo C. Diniz:** Writing – review & editing, Writing – original draft, Investigation. **Fernando V. Lima:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Conceptualization. **Mauro H. Chagas:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Conceptualization. **Hugo C. Martins-Costa:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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